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On-line Estimation in Fed-batch Fermentation Process Using State Space Model and Unscented Kalman Filter*

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Abstract On-line estimation of unmeasurable biological variables is important in fermentation processes, directly influencing the optimal control performance of the fermentation system as well as the quality and yield of the targeted product. In this study, a novel strategy for state estimation of fed-batch fermentation process is proposed. By combining a simple and reliable mechanistic dynamic model with the sample-based regressive measurement model, a state space model is developed. An improved algorithm, swarm energy conservation particle swarm optimization (SECPSO), is presented for the parameter identification in the mechanistic model, and the support vector machines (SVM) method is adopted to establish the nonlinear measurement model. The unscented Kalman filter (UKF) is designed for the state space model to reduce the disturbances of the noises in the fermentation process. The proposed on-line estimation method is demonstrated by the simulation experiments of a penicillin fed-batch fermentation process.
Keywords on-line estimation, simplified mechanistic model, support vector machine, particle swarm optimization, unscented Kalman filter

1 INTRODUCTION

A key problem in fermentation process control is how to monitor the process variables, such as biomass, substrate or product concentrations, in a reliable and cost effective manner. Fermentation processes are generically characterized as complex systems exhibiting non-linear and time-varying dynamics [1]. The optimal control for a fed-batch fermentation system presents a challenge to control engineers because it is difficult to on-line estimate the unmeasurable biological variables. Although many efforts have been made in the development of new hardware sensors and chemical analyzers, commercial reliable and cost effective instruments for on-line measurement are very limited. Hence, some on-line estimation methods, which are based on the mechanistic models or data-driven models, have been proposed for fermentation processes [2].

Many mechanistic models for various fermentation processes have been investigated [3, 4]. These models describe some particular processes and the studies are valuable to simulations and optimization control of fermentation processes. However, it is rather difficult to use the mechanistic model in the on-line estimation for two reasons: (1) the mechanistic model is too complex to construct; (2) the design of nonlinear observer for the mechanistic model is difficult because of large uncertainty of the parameters in the model [5].

The support vector machines (SVM) method, which has solid theoretical foundation rooted in statistical learning theory [6], is regarded as a state-of-the-art technique in the data-driven nonlinear modeling applications. It has been successfully introduced for the soft-sensor applications to estimate the key process variables in fed-batch fermentation processes [7, 8]. However, the estimation accuracy of the SVM method is largely dependent on the data of training set, and the model based on this method is not easy to explain.

Process noise is a common feature of many chemical and biological reactors. Fed-batch fermentation processes are obviously susceptible to environmental effects because of direct interaction through the inlet and outlet streams [9]. Therefore, it is necessary to design nonlinear filters for more accurate estimation. The extended Kalman filter (EKF) is widely used for state estimation and tracking in many engineering applications. However, the use of EKF in a highly nonlinear system leads to some problems. The linearization at each time step may introduce large errors and even cause divergence of the filter. Moreover, calculation for the Jacobian matrices of a high order system is a heavy job for on-line estimation [10]. The unscented Kalman filter (UKF) proposed by Julier and Uhlmann [11] avoids the linearization in the EKF update formula by an unscented transformation (UT) and overcomes the drawbacks of the EKF. The performance of the UKF is better than that of the EKF in terms of robustness and speed of convergence [12].

In this paper, a state space model of a fermentation process is established, in which a simple and reliable mechanistic model is used to predict the nonlinear dynamic equations and the SVM regression is adopted as the measurement equations. The parameters in dynamic equations are identified by swarm energy conservation particle swarm optimization (SECPSO) algorithm. The key state variables are on-line estimated by UKF. The proposed method is applied to a simulated penicillin fed-batch fermentation process.

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2 METHOD

2.1 State space model of fermentation process

The fed-batch fermentation process is a complex non-linear dynamic process. Its dynamic model is typically given by a set of differential equations. A fed-batch fermentation process is described in a state space model as follows

$$\begin{cases} \dfrac{\mathrm{d}\boldsymbol{x}(t)}{\mathrm{d}t} = f(\boldsymbol{x}(t), \boldsymbol{u}(t)) + \boldsymbol{w}(t) \\ \boldsymbol{y}(t) = h(\boldsymbol{x}(t)) + \boldsymbol{v}(t) \end{cases} \tag{1}$$

where $\boldsymbol{x}$ is the state vector, $\boldsymbol{y}$ is the measurement vector, $\boldsymbol{w}$ is the vector of process noise, assumed to be white Gaussian with zero mean and covariance $\boldsymbol{Q}$, $\boldsymbol{v}$ is the vector of measurement noise, assumed to be white Gaussian with zero mean and covariance $\boldsymbol{R}$, t is time, f and h are general functions that represent the process model and measurement model, respectively.

It is difficult to establish an accurate mechanism model for a complex fermentation process. By considering the key state variables, such as culture volume, concentrations of substrate, biomass, and product, and the key control variables such as substrate feed, a simplified mechanistic model can be obtained according to the experiences of experts and the analysis of experiment data. The simplified mechanistic model can express the main characteristics of the fermentation process.

Generally, the relationship between the measurement vector and the state vector can not be described by a direct function. For example, there are no mass conservation equations for dissolved oxygen, carbon dioxide and heat generated with the unmeasured state variables. The measured variables are usually not included in the simplified mechanistic model. Adding more measurements to the system reduces the condition number of the observability matrix and increases the observability degree of state variables [13]. For using more prior knowledge, all available measured variables should be chosen to construct the SVM models. Hence, the SVM method is adopted to supplement the nonlinear measurement equations.

$$\boldsymbol{y} = \mathrm{SVM}(\boldsymbol{x}) \tag{2}$$

where SVM denotes the nonlinear formulation established using SVM model.

For the proposed state space model, the process model is the simplified dynamic mechanistic model. Except for the on-line measurements in the state variables, the SVM measurement equations are adopted. The SVM equations can compensate the non-modeling part of the simplified mechanistic model so that the characteristics of the fermentation process are considered. It is a novel improvement compared with the conventional state space model without SVM equations.

2.2 Parameter identification using SECPSO

The particle swarm optimization (PSO) is an evolutionary computation technique developed by Kennedy and Eberhart [14] based on the simulation of a simplified social model. If there are m particles located in the D-dimensional space to form a community, the position vector of the ith particle is represented as $\boldsymbol{X}_i = [x_{i1}, x_{i2}, \cdots, x_{iD}]^{\mathrm{T}}$, $i = 1, 2, \cdots, m$; and its velocity vector is represented as $\boldsymbol{V}_i = [v_{i1}, v_{i2}, \cdots, v_{iD}]^{\mathrm{T}}$, $i = 1, 2, \cdots, m$. The best position in the searching history of the ith particle is represented as $\boldsymbol{X}_i^*$ and the best position of the particle among the population is represented as $\boldsymbol{X}^*$. For the standard PSO algorithm, in the $(k+1)$ th iteration each particle updates its velocity and position according to the following equations

$$\boldsymbol{V}_i^{k+1} = \omega_1 \boldsymbol{V}_i^k + \eta_1 \mathrm{rand}()\left(\boldsymbol{X}_i^* - \boldsymbol{X}_i^k\right) + \eta_2 \mathrm{rand}()\left(\boldsymbol{X}^* - \boldsymbol{X}_i^k\right) \tag{3}$$

$$\boldsymbol{X}_i^{k+1} = \boldsymbol{X}_i^k + \boldsymbol{V}_i^{k+1} \tag{4}$$

where ω_1 is the inertia weight that determines how much a particle holds its current velocity in the next iteration, η_1 and η_2 are learning factors that control the maximum step size, and rand() is an independent uniformly distributed random variable in the range [0, 1].

In this paper, an improved PSO algorithm called swarm energy conservation PSO (SECPSO) is presented by using the idea of swarm energy conservation. The fitness value J in the SECPSO algorithm is the root mean square error (RMSE) for the simplified mechanistic model values compared with the actual desired values. The fitness value J of the algorithm is calculated according to the following equation

$$J = \sqrt{\frac{\sum_{i=1}^{N}\left(\bar{\boldsymbol{x}}_i - \boldsymbol{x}_i\right)^{\mathrm{T}}\left(\bar{\boldsymbol{x}}_i - \boldsymbol{x}_i\right)}{N}} \tag{5}$$

where $\boldsymbol{x}_i$ is the actual value, $\bar{\boldsymbol{x}}_i$ is the output value of simplified model, and N is the number of samples.

In the standard PSO, a single particle gains experience and information not only from itself but also from the population during its motion. The two types of information are come from the best particle. We attempt to make the failure experience of the worst particle useful. The worst position in the searching history of the ith particle is represented as $\boldsymbol{X}_i^\circ$ and the worst position of the particle among the population is represented as $\boldsymbol{X}^\circ$. In this way, the algorithm updates its velocity equation according to the best value and the worst value of individual and population. The particle can change its status with more information. The update of velocity vector equation with the worst particle is

$$\boldsymbol{V}_i^{k+1} = \omega_1 \boldsymbol{V}_i^k + \eta_1 \mathrm{rand}()\left(\boldsymbol{X}_i^* - \boldsymbol{X}_i^k\right) - \mu_1 \mathrm{rand}()\left(\boldsymbol{X}_i^\circ - \boldsymbol{X}_i^k\right) + \eta_2 \mathrm{rand}()\left(\boldsymbol{X}^* - \boldsymbol{X}_i^k\right) - \mu_2 \mathrm{rand}()\left(\boldsymbol{X}_i^\circ - \boldsymbol{X}_i^k\right) \tag{6}$$

where μ_1 and μ_2 are impact factors, which are used to affect the worst value of individual and population in the updating of the velocity. Other parameters are the same as those in the standard PSO.

The energy of the ith particle is represented as $g_i = \boldsymbol{V}_i^{\mathrm{T}}\boldsymbol{V}_i$. The energy of the swarm is described as

$$G = \sum_{i=1}^{m} g_i = \sum_{i=1}^{m} \boldsymbol{V}_i^{\mathrm{T}}\boldsymbol{V}_i \tag{7}$$

For the system of particle swarm, the energy of the system is conserved. In the other word, the particles that have not fallen into the optimum will get much more energy. Then the probability of obtaining the optimal solution will be increased. The swarm is divided into two sub-swarms: a good sub-swarm and a bad sub-swarm, which are not separated. The particles among them are updated dynamically and good particle may be changed to bad one after each update or *vice versa*.

We set a swarming coefficient $\tau \in [0,1]$ in the SECPSO algorithm. The number of good particles is $\tau \times m$ and that of bad particles is $(1-\tau)\times m$. The particles are arranged according to the fitness value J by ascending order. The $\tau \times m$ particles in the front of the sequence are the good sub-swarm and the rest of particles are the bad sub-swarm. The good particles update their velocity and position according to Eqs. (4) and (6). After the update, the energy of good sub-swarm is represented as

$$G_1 = \sum_{i=1}^{\tau \times m} \boldsymbol{V}_i^{\mathrm{T}}\boldsymbol{V}_i \tag{8}$$

The bad particles update their velocity according to Eq. (3) and their position according to the following equations

$$\boldsymbol{X}_i^{k+1} = \boldsymbol{X}_i^{k} + \delta \boldsymbol{V}_i^{k+1} \tag{9}$$

$$\delta \boldsymbol{V}_{i+1}^{\mathrm{T}}\boldsymbol{V}_{i+1} = \frac{G - G_1}{m(1-\tau)} \tag{10}$$

where δ is the penalty coefficient, representing the difference between the swarm energy G and the good sub-swarm energy G_1, which affects each bad particle uniformly.

The population in the SECPSO algorithm is divided into two sub-swarms dynamically according to the energy of particles. When a bad particle finds the best position, its velocity is zero and it is changed to a good particle. Because of the energy conservation, there must be some other particles, which keep moving in the search space to find more optimal positions locally and globally. Theoretically all the optimal positions can be found as long as the number of particles is large enough. Good particles update their velocity according to the strategy with the worst particle. Bad particles update their velocity according to the strategy with penalty policy. The premature convergence of the standard PSO algorithm can be prevented because the bad sub-swarm undertakes the swarm energy loss generated by velocity reduction of the good sub-swarm. Moreover, the worst position is introduced into the improved PSO and the particles can utilize more information to enhance the cooperation and competition. Therefore, the SECPSO algorithm has faster convergence rate than the standard PSO algorithm at the same iteration step length. Hence, we use the SECPSO algorithm for identifying the parameters of simplified mechanistic equations in the state space model. When the fitness value J reaches the desired value, the optimization procedure stops and the optimal parameters for the simplified model are obtained.

2.3 UKF algorithm

Unlike the EKF, the UKF does not approximate the nonlinear process and measurement model. The UKF uses a minimal set of sample points to capture the true mean and covariance of the nonlinear process. When this set of sample points propagates through the nonlinear system, the posterior mean and covariance are accurately estimated with errors introduced in the second or higher orders. Thus we choose the UKF for the nonlinear state space model.

The following discrete-time state space model is considered, which is from the forward Euler discretization of Eq. (1) and with the same variable declarations.

$$\begin{cases} \boldsymbol{x}(k+1) = f[\boldsymbol{x}(k), \boldsymbol{u}(k)] + \boldsymbol{w}(k) \\ \boldsymbol{y}(k) = h[\boldsymbol{x}(k)] + \boldsymbol{v}(k) \end{cases} \tag{11}$$

A general framework for recursive estimation of states based on the UKF technique is presented as follows [15].

Initialization: ($k = 0$)

$$\hat{\boldsymbol{x}}_0 = \mathrm{E}[\boldsymbol{x}_0] \qquad \boldsymbol{P}_0 = \mathrm{E}\left[(\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_0)(\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_0)^{\mathrm{T}}\right] \tag{12}$$

For $k \in \{1, \cdots, \infty\}$

Step 1: Selection of sigma points. The n-dimensional variable $\boldsymbol{x}_{k-1}$ with mean $\hat{\boldsymbol{x}}_{k-1}$ and covariance $\boldsymbol{P}_{k-1}$ is approximated by sigma points selected

$$\begin{cases} \boldsymbol{\chi}_{i,k-1} = \hat{\boldsymbol{x}}_{k-1}, & i = 0, \\ \boldsymbol{\chi}_{i,k-1} = \hat{\boldsymbol{x}}_{k-1} + \left(\sqrt{(n+\lambda)\boldsymbol{P}_{k-1}}\right)_i, & i = 1, \cdots, n, \\ \boldsymbol{\chi}_{i,k-1} = \hat{\boldsymbol{x}}_{k-1} - \left(\sqrt{(n+\lambda)\boldsymbol{P}_{k-1}}\right)_{i-n}, & i = n+1, \cdots, 2n \end{cases} \tag{13}$$

where $\left(\sqrt{(n+\lambda)\boldsymbol{P}_{k-1}}\right)_i$ is the ith column of the matrix square root of $\sqrt{(n+\lambda)\boldsymbol{P}_{k-1}}$. $\lambda = \alpha^2(n+\kappa) - n$ is the scaling parameter, where constant α determines the spread of the sigma points around $\hat{\boldsymbol{x}}$ and is usually set to $1\times10^{-4} \leqslant \alpha \leqslant 1$, and κ is a secondary scaling parameter and is usually set to 0 for state estimation [11].

Step 2: Prediction. Each point is instantiated through the process model to yield a set of transformed samples

$$\boldsymbol{\chi}_{i,k|k-1} = f\left(\boldsymbol{\chi}_{i,k-1}, \boldsymbol{u}_{k-1}\right), \qquad i = 0,1,\cdots,2n \tag{14}$$

The predicted mean and covariance are computed as

$$\hat{\boldsymbol{x}}_{k|k-1}=\sum_{i=1}^{2n}W_i^{(m)}\boldsymbol{\chi}_{i,k|k-1} \tag{15}$$

$$\boldsymbol{P}_{k|k-1}=\sum_{i=0}^{2n}W_i^{(c)}\left[\boldsymbol{\chi}_{i,k|k-1}-\hat{\boldsymbol{x}}_{k|k-1}\right]\times \left[\boldsymbol{\chi}_{i,k|k-1}-\hat{\boldsymbol{x}}_{k|k-1}\right]^{\mathrm{T}}+\boldsymbol{Q} \tag{16}$$

Step 3: Update. As the measurement equation is nonlinear, the measurement update is performed as

$$\boldsymbol{y}_{i,k|k-1}=h\left(\boldsymbol{\chi}_{i,k|k-1}\right),\qquad i=0,1,\cdots,2n \tag{17}$$

$$\hat{\boldsymbol{y}}_{k|k-1}=\sum_{i=0}^{2n}W_i^{(m)}\boldsymbol{y}_{i,k|k-1} \tag{18}$$

$$\boldsymbol{P}_{y_ky_k}=\sum_{i=0}^{2n}W_i^{(c)}\left[\boldsymbol{y}_{i,k|k-1}-\hat{\boldsymbol{y}}_{k|k-1}\right]\times \left[\boldsymbol{y}_{i,k|k-1}-\hat{\boldsymbol{y}}_{k|k-1}\right]^{\mathrm{T}}+\boldsymbol{R} \tag{19}$$

$$\boldsymbol{P}_{x_ky_k}=\sum_{i=0}^{2n}W_i^{(c)}\left[\boldsymbol{\chi}_{i,k|k-1}-\hat{\boldsymbol{x}}_{k|k-1}\right]\left[\boldsymbol{y}_{i,k|k-1}-\hat{\boldsymbol{y}}_{k|k-1}\right]^{\mathrm{T}} \tag{20}$$

$$\boldsymbol{K}=\boldsymbol{P}_{x_ky_k}\boldsymbol{P}_{y_ky_k}^{-1} \tag{21}$$

$$\hat{\boldsymbol{x}}_k=\hat{\boldsymbol{x}}_{k|k-1}+\boldsymbol{K}\left(\boldsymbol{y}_k-\hat{\boldsymbol{y}}_{k|k-1}\right) \tag{22}$$

$$\boldsymbol{P}_k=\boldsymbol{P}_{k|k-1}-\boldsymbol{K}\boldsymbol{P}_{y_ky_k}\boldsymbol{K}^{\mathrm{T}} \tag{23}$$

Step 4: Repeat of Steps 1 to 3 for the next sample.

In the UKF implementation, the following variable definitions are used

$$\begin{cases}W_0^{(m)}=\lambda/(n+\lambda),\quad W_0^{(c)}=\lambda/(n+\lambda)+\\ \quad(1-\alpha^2+\beta),\quad i=0,\\ W_i^{(m)}=W_i^{(c)}=1/[2(n+\lambda)],\qquad i=1,\cdots,2n\end{cases} \tag{24}$$

where β is used to incorporate prior knowledge of the distribution of $\boldsymbol{x}$. For Gaussian distributions, $\beta=2$ is the optimal [12].

Clearly, the implementation of the UKF is convenient, because it does not need to calculate the Jacobian matrices, while it is necessary in the EKF. That is the advantage of using UKF for the state space model with SVM measurement equations.

3 CASE STUDY AND RESULTS

3.1 Application in fed-batch penicillin fermentation

A fed-batch penicillin fermentation process is taken as an example for application. The complicated fermentation model for penicillin production in a fed-batch fermentor [16] is used as the simulator of the process. The duration of each batch is 400 h, and the sampling interval is chosen to be 0.25 h. White noise is added to CO_2 and dissolved oxygen profiles to account for the sensor sensitivities.

A state space model for the process is established using the proposed modeling method. The priori knowledge data needed in the modeling are all from the batch data of the above simulator. The unmeasured key variables are biomass concentration, production concentration and substrate concentration. These three variables and the culture volume are chosen to construct the simplified mechanistic model for the penicillin fermentation. The measurement equations are constructed by using four measured variables: dissolved oxygen, carbon dioxide, heat generated and culture volume. The state space model involves the dynamic equations and the measurement equations.

The dynamic equations are as follows

$$\begin{cases}\dfrac{\mathrm{d}V_l}{\mathrm{d}t}=F-F_{\mathrm{loss}}\\ \dfrac{\mathrm{d}C_{\mathrm{x}}}{\mathrm{d}t}=\mu C_{\mathrm{x}}-\dfrac{C_{\mathrm{x}}}{V_l}\dfrac{\mathrm{d}V_l}{\mathrm{d}t}\\ \dfrac{\mathrm{d}C_{\mathrm{p}}}{\mathrm{d}t}=\mu_{\mathrm{p}}C_{\mathrm{x}}-K_{\mathrm{d}}C_{\mathrm{p}}-\dfrac{C_{\mathrm{p}}}{V_l}\dfrac{\mathrm{d}V_l}{\mathrm{d}t}\\ \dfrac{\mathrm{d}C_{\mathrm{s}}}{\mathrm{d}t}=-\left(\dfrac{\mu}{Y_{\mathrm{x/s}}}+\dfrac{\mu_{\mathrm{p}}}{Y_{\mathrm{p/s}}}+Q_{\mathrm{m}}\right)C_{\mathrm{x}}+\dfrac{FS_0}{V_l}-\dfrac{C_{\mathrm{s}}}{V_l}\dfrac{\mathrm{d}V_l}{\mathrm{d}t}\\ F_{\mathrm{loss}}=V_l\lambda_{\mathrm{loss}}\left\{\mathrm{e}^{5[(T-T_0)/(T_{\mathrm{c}}-T_0)]}-1\right\}\\ \mu=\mu_{\max}\dfrac{C_{\mathrm{s}}}{K_{\mathrm{s}}C_{\mathrm{x}}+C_{\mathrm{s}}}\\ \mu_{\mathrm{p}}=\mu_{\mathrm{pmax}}\dfrac{C_{\mathrm{s}}}{K_{\mathrm{i}}+C_{\mathrm{s}}+C_{\mathrm{s}}^2/K_{\mathrm{p}}}\end{cases} \tag{25}$$

The measurement equations are

$$\begin{cases}C_{\mathrm{o}}=\mathrm{SVM}_{\mathrm{o}}\left(C_{\mathrm{x}},C_{\mathrm{s}},C_{\mathrm{p}},V_l\right)\\ C_{\mathrm{c}}=\mathrm{SVM}_{\mathrm{c}}\left(C_{\mathrm{x}},C_{\mathrm{s}},C_{\mathrm{p}},V_l\right)\\ Q_{\mathrm{heat}}=\mathrm{SVM}_{\mathrm{heat}}\left(C_{\mathrm{x}},C_{\mathrm{s}},C_{\mathrm{p}},V_l\right)\\ V_l=V_l\end{cases} \tag{26}$$

where the measurement vector is $\boldsymbol{y}=\left[C_{\mathrm{o}},C_{\mathrm{c}},Q_{\mathrm{heat}},V_l\right]^{\mathrm{T}}$. The discrete-time state space model is obtained from forward Euler discretization of Eqs. (25) and (26). The discrete-time interval is 0.25 h. One batch data of the simulator are chosen as the priori knowledge data and the parameter values of dynamic equations are estimated using the SECPSO algorithm. The SVM models in measurement equations are established using ε-SVR (support vector regression) and the batch data from the penicillin simulator are taken as the training set.

One popular choice of parameters is $\omega_1=0.7298$, $\eta_1=\eta_2=1.4962$ [17]. The ratio between good particles and bad particles is decided by the swarming coefficient τ. If τ is close to 0, the bad particles are in the majority and the penalty coefficient on their velocity is small. The improvement for the standard PSO is not obvious. If τ is close to 1, the number of bad particles is very limited and their penalty coefficient will be too

Table 1 Values of the parameters obtained using SECPSO

K_i	K_s	K_p	K_d	λ_{loss}	$Y_{x/s}$	$Y_{p/s}$	μ_{max}	μ_{pmax}	Q_m
1.838×10^{-11}	0.2027	0.4855	0.0146	0.0001	0.4712	0.8318	0.1051	0.0017	0.0171

large, which also leads to worse performance of the SECPSO. The value of swarming coefficient τ can be obtained by experience and experimental simulations. The effect of the worst particle of individual and population in the updating of the velocity is controlled by the impact factors μ_1 and μ_2. If their values are too large, the worst information will disturb the effect of the best information, but if their values are too small, the effect of worst information will be insignificant. The suitable setting strategy for μ_1 and μ_2 is that their values increase linearly with the present iteration number k. According to the results of simulations, we set $\tau = 0.5$ and $\mu_1 = \mu_2 = 0.001k / \text{MaxIN}$, where MaxIN is the maximum iteration number. The values of the parameters obtained using the SECPSO algorithm for the penicillin state space model are listed in Table 1.

A SVM implementation for regression known as the "ε-SVR" in the LIBSVM software library [18] is employed for constructing the SVM models in measurement equations. Three parameters need to be set: C, γ and ε, with C the penalty parameter for the error term, γ as the kernel parameter and ε the precision parameter in loss function of ε-SVR. The value of ε can inversely affect the number of support vectors used to construct the regression function. As ε decreases, the percentage of training points marked as SVs (support vectors) increases, which in turn enhances the complexity of the SVR model. On the other hand, a relatively better generalization performance is realized for large magnitude of ε at the risk of obtaining a high training set error [7]. The value of ε is proportional to the input noise level and its range is usually from 0.001 to 0.1. SVM models are developed for three ε values (0.1, 0.01 and 0.001). The number of SVs for the SVR models increases as ε decreases. The ε value is set according to the number of SVs, which is more close to the half of the training set, a compromise between the generalization and training accuracy. We use a grid search and υ-fold cross-validation method [19] to find optimal parameters (C and γ) for the SVM model. In this simulation, a grid search on C and γ is performed on the training set using a 5-fold cross-validation. Our implementation is as follows. The grid search is realized by evaluating exponential sequences of C and γ (*i.e.*, $C = 2^{-10}, 2^{-9}, \cdots, 2^{10}$; $\gamma = 2^{-10}, 2^{-9}, \cdots, 2^{10}$). Then, all grid points of C and γ are tried to see which one gives the highest cross validation accuracy. We use the best parameters to train the whole training set and generate the final SVM model. Finally, the optimum parameters [C, γ, ε] of SVM model for C_o are [128, 2, 0.01], those for C_c are [4, 2^{-6}, 0.01], and those for Q_{heat} are [64, 2^{-10}, 0.001].

3.2 Experiments and results

For the practical experiment, the initial value for the state vector is unknown and the measurement values are contaminated by the noises. The UKF is designed for the state tracking and estimation of the state space model. We set $\kappa = 0$ and $\beta = 2$. The constant α is sensitive to the filter results and its value can be decided by simulation results. The initial values of $\hat{\boldsymbol{x}}_0$ are set to $[0.001, 14.7, 0.1, 102]^T$, which are close to the true initial values $[0.1, 15, 0, 100]^T$. The initialization tuning parameters, the process noise covariance $\boldsymbol{Q}$ and measurement noise covariance $\boldsymbol{R}$ for the UKF are as follows

$$\boldsymbol{P}_0 = \mathrm{E}\left[\left(\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_0\right)\left(\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_0\right)^{\mathrm{T}}\right] = \mathrm{diag}\left(\left[10^{-7}, 10^{-7}, 10^{-7}, 10^{-6}\right]\right) \tag{27}$$

$$\boldsymbol{Q} = \mathrm{diag}\left(\left[10^{-7}, 10^{-7}, 10^{-7}, 10^{-6}\right]\right) \tag{28}$$

$$\boldsymbol{R} = \mathrm{diag}\left(\left[10^{-3}, 10^{-4}, 10^{-6}, 10^{-6}\right]\right) \tag{29}$$

White noises added on CO_2 and dissolved oxygen profiles are the main noise sources for the state space system. The proper initial tuning parameters for UKF are evaluated by comparison with experimental data through a trial-and-error procedure. We set $\alpha = 1\times10^{-3}$ for the state space model with the SVM equations. The UKF for the state space model without the SVM measurement equations is also designed for comparing the tracking and estimation performance of the proposed on-line estimation method. Both state space models have the same simplified mechanistic model of the penicillin fermentation. The measurement vector for the state space model without the SVM equations is the culture volume. The measurement equation is

$$y = V_l \tag{30}$$

We set $\alpha = 1$ and $\boldsymbol{R} = \mathrm{diag}([10^{-6}])$ for the state space model without the SVM equations. The other tuning parameters have the same values as those with the UKF of the state space model with the SVM equations.

For the simplified mechanistic model, besides the culture volume, the other state vector variables, such as biomass concentration, substrate concentrate and production concentration, are unmeasurable. These state variables can be on-line estimated using the UKF. The results of the on-line estimation using different state space model are shown in Figs. 1−3. The RMSE values for the state variables of the state space model with and without SVM equations compared with the simulator true values are listed in Table 2. For the state space model without SVM equations, except for the culture volume that can track the simulator model values well, the other state variables have large RMSE compared with the simulator values. For the state space with SVM equations, the RMSE for all four

Table 2 RMSE values for different state space models compared with the simulator true values

Modeling formalism	RMSE(C_x)	RMSE(C_s)	RMSE(C_p)	RMSE(V_l)
without SVM equations	2.8180	6.9591	0.2072	0.1381
with SVM equations	0.2308	0.0958	0.0413	0.1351

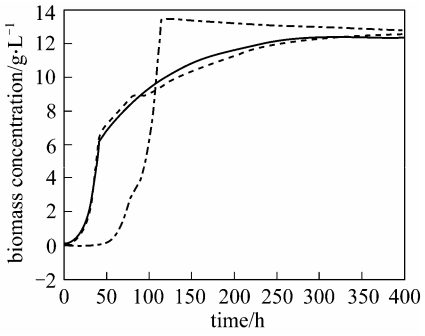


Figure 1 Comparison of biomass concentration for different state space models
—— simulator; ---- with SVM; - - - - without SVM

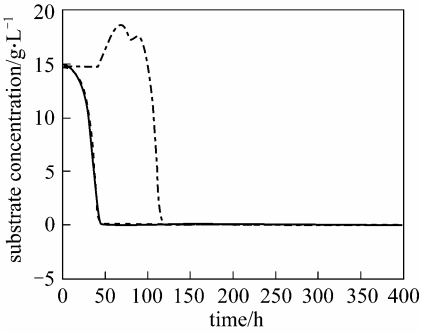


Figure 2 Comparison of substrate concentration for different models
—— simulator; ---- with SVM; - - - - without SVM

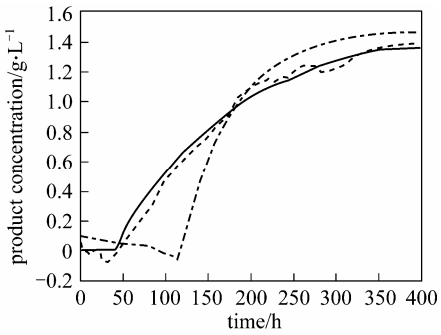


Figure 3 Comparison of product concentration for different models
—— simulator; ---- with SVM; - - - - without SVM

state variables is small and the UKF is robust for the initial values. The proposed on-line estimation method can track the simulator values well and it is effective for the on-line estimation in fed-batch fermentation process.

4 CONCLUSIONS

The proposed estimation method improves the accuracy of on-line estimation in a fed-batch fermentation process. The experiments, carried out using the data from the simulated penicillin fermentation, have validated the main characteristics of the proposed approach. The tracking and estimation performance of the UKF for the state space model with SVM measurement equations is compared with that using the mechanistic state model only. The results indicate that the proposed method is an effective approach to address the difficult on-line estimation of fed-batch fermentation processes.

NOMENCLATURE

C	penalty parameter for the error term in SVM model
C_i	concentration of component i, g·L^{-1}
D	dimension of particles in SECPSO algorithm
F	feed flow rate, L·h^{-1}
F_{loss}	evaporative loss rate, L·h^{-1}
f	function of process model
G	swarm energy in SECPSO algorithm
G_1	good sub-swarm energy in SECPSO algorithm
g	particle energy in SECPSO algorithm
h	function of measurement model
J	fitness value in SECPSO algorithm
$\boldsymbol{K}$	Kalman gain matrix
K_d	production hydrolysis rate constant, h^{-1}
K_i	inhibition constant, g·L^{-1}
K_p	inhibition constant for product formation, g·L^{-1}
K_s	saturation constant for substrate (glucose)
k	present iteration number in SECPSO algorithm
m	number of particles in SECPSO algorithm
N	number of samples
n	dimension of state vector
$\boldsymbol{P}$	covariance of state vector
$\boldsymbol{Q}$	covariance of process noise
Q_{heat}	heat generation, J
Q_m	glucose consumption rate for maintenance energy, g·g^{-1}·h^{-1}
$\boldsymbol{R}$	covariance of measurement noise
S_0	substrate concentration of feed, g·L^{-1}
T	culture temperature (25 °C)
T_e	boiling temperature (100 °C)
T_0	freezing temperature (0 °C)
t	time, h
$\boldsymbol{u}$	control input vector
$\boldsymbol{V}$	velocity vector of particle in SECPSO algorithm
V_l	volume, L
$\boldsymbol{v}$	vector of measurement noise
W	coefficient in UKF algorithm
$\boldsymbol{w}$	vector of process noise
$\boldsymbol{X}$	position vector of particle in SECPSO algorithm
$\boldsymbol{X}^*$	the best position of the particle among the population in SECPSO algorithm

$\boldsymbol{X}_i^*$ the best position in the searching history of the ith particle in SECPSO algorithm
$\boldsymbol{X}^\circ$ the worst position of the particle among the population in SECPSO algorithm
$\boldsymbol{X}_i^\circ$ the worst position in the searching history of the ith particle in SECPSO algorithm
$\boldsymbol{x}$ state vector
$\bar{\boldsymbol{x}}$ output value of simplified model
$\hat{\boldsymbol{x}}$ estimation value of state vector
$Y_{i/j}$ yield of component i on j, g·g^{-1}
$\boldsymbol{y}$ measurement vector
α constant in UKF algorithm
β scaling parameter used to incorporate prior knowledge of distribution of $\boldsymbol{x}$
γ kernel parameter of RBF in SVM model
δ penalty coefficient in SECPSO algorithm
ε precision parameter in loss function in SVM model
η_1,η_2 learning factors in SECPSO algorithm
κ secondary scaling parameter in UKF algorithm
λ scaling parameter in UKF algorithm
λ_{loss} constant in F_{loss}
μ specific growth rate, h^{-1}
$\mu_{\max}$ maximum specific growth rate, h^{-1}
μ_{p} specific rate of production, h^{-1}
μ_{pmax} maximum specific rate of production, h^{-1}
μ_1,μ_2 impact factors in SECPSO algorithm
τ swarming coefficient in SECPSO algorithm
$\boldsymbol{\chi}$ sigma points in UKF algorithm
ω_1 inertia weight in SECPSO algorithm

Subscripts

c carbon dioxide
o dissolved oxygen
p product
s substrate (glucose)
x biomass

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